# Structured Data on News & Fact-Checking Domains: Extraction and Quality Analysis of WDC Schema.org Annotations

Technical Report

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# 1 Motivation and Scope

Structured semantic annotations embedded in web pages allow publishers to represent page content as machine-readable triples ⟨subject, predicate, object⟩ using shared vocabularies such as Schema.org [1]. Over 51% of pages in the Common Crawl web archive already carry such annotations [2], yet their quality remains limited: most Schema.org properties are left unpopulated, and annotations tend to describe page metadata rather than actual content.

This work examines that gap specifically for news outlets and fact-checking organisations, sources that are particularly relevant for automated fact-checking. The analysis is intended to motivate and ground a broader research on using large language models (LLMs) to automatically extract richer, content-grounded annotations from web pages, whose output could serve as structured evidence for fact-checking systems.

In this work, we assess the completeness, syntactic correctness, structural complexity, and internal consistency of Schema.org annotations across a curated set of news and fact-checking domains, and conduct a dedicated analysis of the `ClaimReview` [3] class used by fact-checking organisations.

The **Web Data Commons (WDC)** project [4] (`https://webdatacommons.org`) extracts all structured data embedded in the Common Crawl web archive [5] and publishes it as N-Quad files. The 2024-12 release is the latest snapshot and covers the entire crawl; it also provides 50 Schema.org class-specific subsets, each includes all quads whose source pages declare at least one entity of that class.

This report documents the full pipeline used to retrieve, process, and analyse WDC data restricted to a curated list of news and fact-checking domains.

# 2 WDC Data Retrieval Strategy

## 2.1 Sources

Targeted domains were assembled from two primary sources:

**Fact-checking domains.** The **Fact-Check Insights** dataset [6] (`https://factcheckinsights.org/`) is a compilation of all fact-checks published by organisations verified by the Duke Reporters' Lab that carry `ClaimReview` and/or `MediaReview` markup. We extracted the fact-checking organisation domains from `author.url` field of `claim_review.csv` we ranked them by publication frequency and we selected the top 15.

**News domains.** The **Media Cloud Directory API** [7] (`https://www.mediacloud.org`) indexes over 60,000 news sources worldwide. National collections for each country were filtered to the top 5th percentile by total story count and stories per week, yielding high-volume national sources from every country.

The two collections were merged into a final list of **388 domains**, which forms the basis for the retrieval pipeline described in the next subsections.

## 2.2 The PLD Lookup Table Approach

For each of the 50 WDC class-specific subsets, WDC publishes a **PLD lookup CSV** that maps pay-level domains to the chunk file numbers that contain their data. This makes it possible to have a proper sample without downloading the full corpus: for each class, only the chunks that include quads of the target domain list need to be retrieved.

1. Download the lookup CSV for every class.

2. Intersect each lookup with the 388 target domains.
3. Record which chunks are needed for which class.
4. Download and filter only those chunks.

Out of 50 classes, **37 had at least one matching domain**; 13 had none.

### 2.3 Priority-Based Chunk Reduction

The naive plan required downloading **1,093 chunks** across 38 classes, estimated at roughly one week of sequential downloads. A priority filter reduced the scope to **134 chunks** (37 classes) by applying the following rules:

- At most **5 chunks per class** are selected.
- Chunks are ranked by a score combining the count of matching target PLDs they contain and their type: **core PLDs** include curated high-trust domains such as major international news outlets (`reuters.com`, `apnews.com`, `bbc.com`, `aljazeera.com`, `dw.com`, `theguardian.com`, `nytimes.com`, `thetimes.com`, `lemonde.fr`, `cnn.com`), `en.wikipedia.org`, and dedicated fact-checkers (`politifact.com`, `snopes.com`, `factcheck.org`, `fullfact.org`). These domains contribute a higher weight to the score than general news domains, so chunks covering core sources are preferred.
- **Core PLDs** (fact-checkers and major news outlets) bypass the 5-chunk cap: any chunk containing a core PLD is always included.
- Each core PLD is allowed to appear in at most 5 classes to prevent excessive duplication across the merged output.

The result is an **88% reduction** in download volume (1,093 → 134 chunks) with full representation of the most important domains in each class.

## 3 Extraction Results

### 3.1 Per-Class Output

Table 1 summarises the extraction statistics for all 37 classes that yielded data. Total raw matches (before global deduplication) sum to **78,940,582 quads**.

Table 1: Per-class extraction results. Quads = lines in the per-class `.nq` quads file (before cross-class deduplication).

| Class | Quads | File (MB) | Chunks |
|---|---|---|---|
| Country | 26,683,479 | 4,409.1 | 5 |
| SportsEvent | 9,633,393 | 1,754.4 | 4 |
| Place | 7,807,918 | 1,580.6 | 4 |
| AdministrativeArea | 6,752,407 | 1,613.8 | 4 |
| SportsTeam | 5,786,149 | 1,019.1 | 4 |
| Event | 5,380,763 | 1,163.7 | 5 |
| GeoCoordinates | 4,070,109 | 971.3 | 5 |
| LocalBusiness | 3,489,727 | 609.8 | 5 |
| Person | 2,433,147 | 621.6 | 7 |

Table 1 (continued)

| Class | Quads | File (MB) | Chunks |
|---|---|---|---|
| Organization | 2,150,703 | 416.4 | 6 |
| Product | 1,763,672 | 502.9 | 4 |
| QAPage | 1,313,529 | 337.0 | 5 |
| ClaimReview | 470,994 | 115.4 | 1 |
| Language | 372,426 | 91.1 | 5 |
| SearchAction | 294,263 | 73.2 | 5 |
| Answer | 265,441 | 73.7 | 5 |
| Question | 237,012 | 68.7 | 5 |
| FAQPage | 218,073 | 65.4 | 5 |
| Hospital | 243,676 | 34.5 | 1 |
| Recipe | 207,258 | 42.6 | 5 |
| JobPosting | 719,570 | 160.6 | 5 |
| Dataset | 141,510 | 22.9 | 1 |
| Movie | 143,037 | 26.9 | 5 |
| StadiumOrArena | 736,013 | 128.4 | 2 |
| City | 763,941 | 141.0 | 3 |
| TVEpisode | 111,867 | 16.7 | 2 |
| RadioStation | 67,173 | 13.5 | 1 |
| Restaurant | 48,786 | 11.4 | 5 |
| Book | 30,572 | 7.6 | 5 |
| MusicRecording | 19,070 | 3.7 | 1 |
| Hotel | 4,281 | 0.8 | 1 |
| ShoppingCenter | 3,599 | 0.7 | 1 |
| CollegeOrUniversity | 2,198 | 0.6 | 2 |
| GovernmentOrganization | 1,517 | 0.3 | 1 |
| LandmarksOrHistoricalBuildings | 624 | 0.2 | 1 |
| CreativeWork | 87,243 | 18.2 | 5 |
| EducationalOrganization | 252 | 0.06 | 2 |
| **Total** | **78,940,582** | **16,505** | **134** |

Note that these figures do not correspond exclusively to entities of the listed classes. Rather, each class-specific dataset includes all quads extracted from pages where that class occurs, including annotations of other entity types present on the same pages.

### 3.2 Merging and Deduplication

All 37 per-class `.nq` files were merged into `output/merged.nq` with **global exact-line deduplication** across classes.

Table 2: Merged corpus summary.

| Metric | Value |
|---|---|
| Unique quads (lines in `merged.nq`) | 69,621,246 |
| Uncompressed file size | 14,360 MB (14.4 GB) |
| Unique registered domains | 119 |
| Unique subdomains (hosts) | 363 |
| Distinct Schema.org types declared | 162 |

Cross-class deduplication removed approximately 11.8% of raw quads (9.3M), confirming that many pages appear in multiple per-class extracts. The 119 unique registered domains recovered (out of 388 targeted) indicates that a substantial share of the target list has no WDC footprint, primarily because many news sources block Common Crawl or were not crawled in the 2024-12 snapshot.

### 3.3 Domain Coverage

Table 3 lists the top 15 registered domains in the merged corpus by quad count, combining page coverage, annotation density, and entity statistics for each outlet.

Table 3: Top 15 registered domains by quad count (merged corpus). *Pages*: distinct crawled pages; *Quads*: total RDF quads extracted; *%Q*: share of the 69.6 M corpus quads; *Q/p*: quads per page (annotation density); *Entities*: distinct entities across the domain's pages; *%E*: share of the ∼12.2 M corpus entities.

| **Domain** | **Pages** | **Quads** | **%Q** | **Q/p** | **Entities** | **%E** |
|---|---|---|---|---|---|---|
| ria.ru | 101,258 | 26,812,929 | 38.5 | 264.8 | 3,638,349 | 29.8 |
| elpais.com | 36,821 | 7,975,385 | 11.5 | 216.6 | 1,301,395 | 10.7 |
| cbssports.com | 17,983 | 5,117,711 | 7.4 | 284.6 | 1,221,854 | 10.0 |
| sports.ru | 38,174 | 3,043,424 | 4.4 | 79.7 | 723,710 | 5.9 |
| freepressjournal.in | 15,986 | 2,800,707 | 4.0 | 175.2 | 675,231 | 5.5 |
| indiatimes.com | 33,332 | 2,524,858 | 3.6 | 75.7 | 503,533 | 4.1 |
| rbc.ua | 30,542 | 2,455,246 | 3.5 | 80.4 | 408,279 | 3.3 |
| cnn.com | 18,825 | 1,915,302 | 2.8 | 101.7 | 281,225 | 2.3 |
| cbsnews.com | 12,334 | 1,700,837 | 2.4 | 137.9 | 251,695 | 2.1 |
| lefigaro.fr | 24,615 | 1,533,654 | 2.2 | 62.3 | 342,138 | 2.8 |
| as.com | 17,678 | 1,505,213 | 2.2 | 85.1 | 340,609 | 2.8 |
| elespanol.com | 13,306 | 1,284,796 | 1.8 | 96.6 | 248,921 | 2.0 |
| aif.ru | 30,720 | 1,241,772 | 1.8 | 40.4 | 266,521 | 2.2 |
| wikipedia.org | 66,735 | 1,228,468 | 1.8 | 18.4 | 277,768 | 2.3 |
| dw.com | 41,245 | 826,324 | 1.2 | 20.0 | 208,339 | 1.7 |
| *Remaining 104 domains* | *each <1.2% of total quads* | | | | | |

The appendix provides the complete list of domains. It is worth noting that some news sources block the Common Crawl bot from indexing their pages, so not all domains are expected to appear in the retrieved data. Nevertheless, many major and high-value sources are well-represented. The top domain (`ria.ru`) accounts for 38.5% of the corpus by quad count, introducing a skew towards the Russian source. However, the remaining 61.5% includes a highly diverse collection of domains, including some highly trusted global ones (e.g. elpais, cnn, dw, cbsnews) Annotation density (Q/p) ranges from 18.4 (`wikipedia.org`) to 284.6 (`cbssports.com`), reflecting differences in how many distinct Schema.org entities each publisher embeds per page.

### 3.4 Schema.org Type Distribution

The merged corpus contains 162 distinct Schema.org types. The top 20 by entity count are shown in Figure 1.

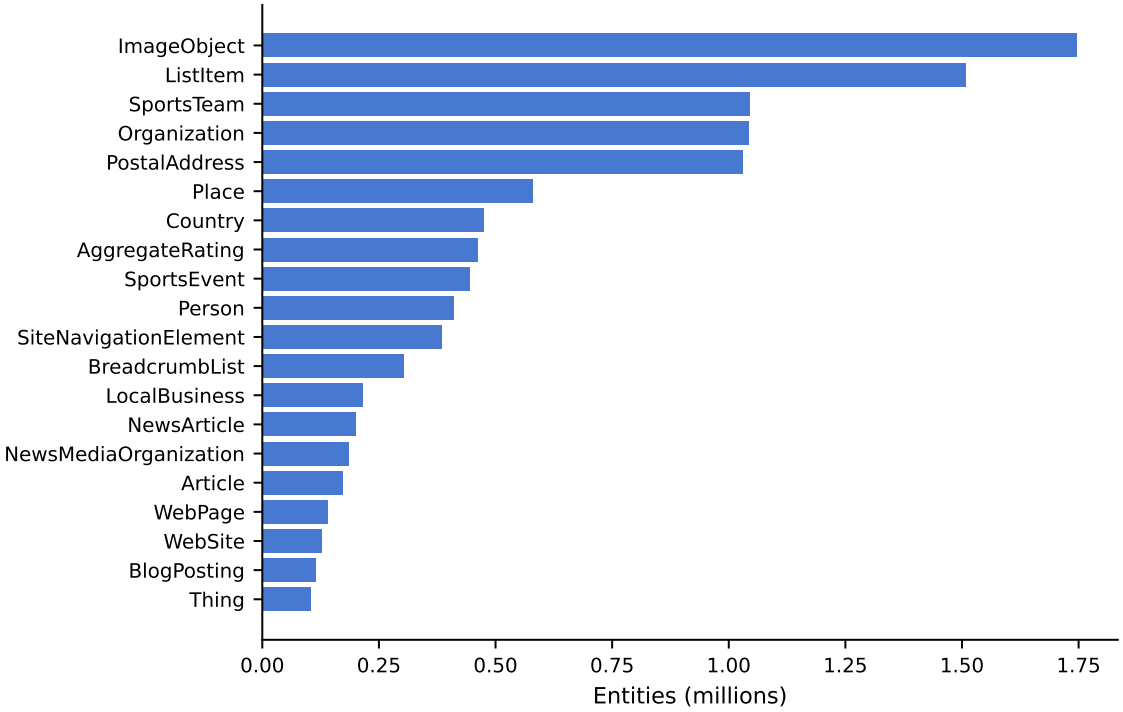



Figure 1: Top 20 declared entity types in the merged corpus, sorted by entity count.

The top 20 types are dominated by structural and presentational classes: `ImageObject`, `ListItem`, `BreadcrumbList`, and `SiteNavigationElement` are generated automatically by CMS templates and carry no article-level semantic content. `NewsArticle` (199,809) and `Article` (172,671) are the content-primary types that interest us in the top 20, yet together they represent under 3% of total entities, illustrating how thin the content-grounded annotation layer is relative to the surrounding structural markup. A dedicated analysis of the article and news article pages is conducted in 4.11.

In terms of usage across domains, types such as `Article`, `NewsArticle`, `NewsMediaOrganization`, `Person`, and `Place` appear in many domains, while the remaining types vary considerably in their distribution, with some concentrated in specific sources and others present in only a small number of entities corpus-wide.

# 4 Full-Corpus Quality Analysis

## 4.1 From N-Quads to DuckDB

Applying the in-memory entity-store approach to the full 14 GB merged file caused an out-of-memory error. The solution switches to a DuckDB [8] columnar store:

1. The 14 GB file is read **line by line** in a single Python process.
2. Every 1 000 000 lines, four Python lists are flushed to a ∼100 MB Parquet file and immediately cleared.
3. After all ∼70 Parquet files are written, DuckDB bulk-loads them using its own internal streaming; no large Python object is ever held in RAM.

Table 4: DuckDB entity store statistics.

| Metric | Value |
|---|---|
| Triples ingested | 69,600,381 |
| Parse errors (skipped) | 20,865 (0.03%) |
| Entities | 12,211,353 |
| DuckDB store size | 18.61 GB |

The 20,865 parse errors (0.03%) are malformed N-Quad lines falling into two categories: **20,863** lines from `navbharattimes.indiatimes.com` carry language-tagged literals with an unfilled template placeholder (`"Hindi News"@$lang`), where the site's JSON-LD renderer fails to substitute `$lang` with a valid BCP 47 language tag; and **2** lines from `politifact.com` embed an HTML fragment inside a URI field (e.g., `<https://schema.org/url</div>>`), making the predicate unparseable.

### 4.2 Namespace Correctness

Before assessing semantic content, the technical correctness of Schema.org namespace URIs is verified. Non-canonical or malformed namespaces may render annotations invisible to parsers that perform strict URI matching against the canonical `schema.org` prefix. In practice, however, many production systems apply normalization or tolerate minor variations (e.g., `http` vs `https`), so the impact depends on the specific parser.

Table 5: Schema.org namespace usage in the merged corpus.

| URI class | Count | % |
|---|---|---|
| Canonical (`http://schema.org/` or `https://schema.org/`) | 56,962,451 | 99.90% |
| Non-canonical (`https://www.schema.org/`) | 57,157 | 0.10% |
| Truly malformed | 0 | 0.00% |
| Other namespaces (`rdf:`, `owl:`, etc.) | 405,175 | – |

The top non-canonical example is `https://www.schema.org/SiteNavigationElement` (17,372 occurrences); schema.org does not use the `www.` subdomain. However, the 0.10% non-canonical rate poses negligible analysis risk.

### 4.3 Type Validity

Of all `rdf:type` assertions referencing Schema.org, **99.76%** use a class name that exists in the schema.org vocabulary (979 known classes). Invalid types account for 28,102 declarations (0.24%), predominantly capitalisation errors or retired class names. Table 6 lists the top invalid type names.

Table 6: Top invalid `rdf:type` names by declaration count.

| Invalid type | Declarations |
|---|---|
| imageObject | 16,802 |
| PERSON | 6,286 |
| person | 2,500 |
| Website | 1,015 |
| place | 526 |
| Webpage | 458 |
| liveBlogPosting | 319 |
| webpage | 161 |
| NewsPaper | 15 |
| Expert | 7 |

The top invalid types are simple capitalisation variants of standard classes (`imageObject`, `person`, `place`). These affect only 0.24% of type declarations and do not significantly affect the corpus-level quality assessment.

### 4.4 Property Validity

Property validity is assessed by checking whether each schema.org property is actually defined for the declaring entity's type (using `domainIncludes` with inheritance).

Table 7 lists the 20 most frequently used valid Schema.org properties across the entire corpus (44,753,999 valid uses total).

Table 7: Top 20 most-used valid Schema.org properties.

| Property | Uses |
|---|---|
| name | 6,578,093 |
| url | 4,920,617 |
| height | 1,644,399 |
| width | 1,615,246 |
| position | 1,512,667 |
| image | 1,421,062 |
| address | 1,283,189 |
| description | 1,179,263 |
| contentUrl | 1,104,876 |
| item | 1,017,368 |
| addressLocality | 1,005,532 |
| addressRegion | 945,432 |
| logo | 918,988 |
| addressCountry | 807,141 |
| postalCode | 761,799 |
| streetAddress | 755,247 |
| telephone | 728,709 |
| sameAs | 672,483 |
| datePublished | 616,006 |
| dateModified | 613,182 |

The top properties are dominated by identifiers (`name`, `url`), media dimensions (`height`, `width`), and address fields, all of which serve page presentation or search-engine indexing rather

than deep content annotation. Properties with direct relevance to article content or fact-checking (`articleBody`, `mentions`, `claimReviewed`) do not appear in the top 20, consistent with the metadata-first annotation pattern observed throughout this report.

Table 8: Property validity breakdown.

| **Category** | **Count** | **%** |
|---|---|---|
| Valid property uses | 44,753,999 | 99.18% |
| Invalid property uses | 370,086 | 0.82% |
| *Breakdown of invalid uses by root cause:* | | |
| Untyped entity (no `rdf:type`) | 115,149 | 31.1% |
| Unknown entity type | 101,467 | 27.4% |
| Unknown property (not in vocab) | 93,779 | 25.3% |
| Domain mismatch (wrong type) | 59,691 | 16.1% |

Property violations are classified into four mutually exclusive categories. An **untyped entity** violation occurs when a triple's subject carries no `rdf:type` declaration, making it impossible to verify whether the property is applicable. An **unknown type** violation arises when the declared entity type is not recognised in the Schema.org vocabulary, so no valid property set can be derived. An **unknown property** violation is recorded when the property name itself does not exist in the Schema.org vocabulary, regardless of the entity type. Finally, a **domain mismatch** violation occurs when both the entity type and the property are individually valid Schema.org terms, but the property's `domainIncludes` constraint (including inherited types) does not cover the declared entity type.

The top-20 individual invalid property names broken down by violation reason are shown in Table 9.

Table 9: Top 20 invalid property names by total misuse count, broken down by violation reason.

| **Property** | **Total** | **Untyped** | **Unk. type** | **Unk. prop** | **Dom. mismatch** |
|---|---|---|---|---|---|
| url | 69,754 | 52,232 | 17,522 | 0 | 0 |
| name | 69,619 | 59,167 | 10,452 | 0 | 0 |
| price | 48,464 | 98 | 0 | 0 | 48,366 |
| query-input | 24,424 | 0 | 0 | 24,424 | 0 |
| width | 19,419 | 0 | 16,802 | 0 | 2,617 |
| description | 19,219 | 12 | 19,207 | 0 | 0 |
| postID | 17,323 | 0 | 0 | 17,323 | 0 |
| height | 16,804 | 0 | 16,802 | 0 | 2 |
| Telephone | 16,022 | 0 | 0 | 16,022 | 0 |
| caption | 16,017 | 0 | 16,017 | 0 | 0 |
| partner | 15,085 | 0 | 0 | 15,085 | 0 |
| @name | 9,029 | 1,701 | 0 | 7,328 | 0 |
| primaryImageOfPage | 3,896 | 0 | 0 | 0 | 3,896 |
| SameAs | 3,667 | 0 | 0 | 3,667 | 0 |
| jobTitle | 1,820 | 0 | 7 | 0 | 1,813 |
| haspart | 1,596 | 0 | 0 | 1,596 | 0 |
| publisher | 1,120 | 12 | 323 | 0 | 785 |
| address | 1,094 | 412 | 526 | 0 | 156 |
| isAccessibleForFree | 896 | 0 | 896 | 0 | 0 |
| potentialAction | 896 | 0 | 896 | 0 | 0 |

### 4.5 Range Violations

Range violations occur when a property's value does not conform to the expected type specified by its `rangeIncludes` constraint in the Schema.org vocabulary. For example, a property that expects a `URL` or `Text` literal may instead receive a nested entity, or a property expecting an `Integer` may receive a string. Unlike domain violations, which concern the *subject* of a triple, range violations concern the *object*: the value provided is incompatible with what the vocabulary prescribes for that property.

Table 10: Top range-violating properties (strict semantic type mismatches).

| Property | Violations | Example values |
|---|---|---|
| `height` | 1,640,370 | `"60"`, `"675"` (string instead of numeric/QuantitativeValue) |
| `width` | 1,613,832 | `"600"`, `"300"` (string instead of numeric/QuantitativeValue) |
| `endDate` | 395,515 | malformed or empty date strings |
| `startDate` | 382,002 | non-ISO 8601 date strings |
| `eventStatus` | 188,068 | malformed IRI (missing scheme or invalid format) |
| `eventAttendanceMode` | 158,336 | non-IRI string instead of controlled vocabulary term |
| `datePublished` | 83,558 | non-ISO date strings |
| `dateModified` | 60,531 | non-ISO date strings |
| Total checked: 55,056,887 assertions | | Total violations: 6,054,576 (11.00%) |

The two largest violating properties (`height`, `width`) account for 54% of all violations: publishers provide raw integers where Schema.org expects a `QuantitativeValue` or `Distance` node.

### 4.6 Distinct Schema.org Types per Page

Across the 596,604 pages in the corpus, the mean number of distinct Schema.org types declared is **7.01** (median 7.0, max 29). Figure 2 shows the distribution.

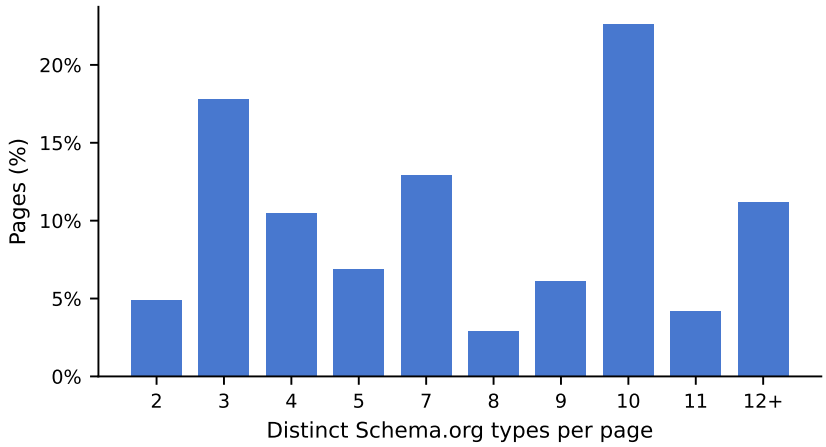



Figure 2: Distribution of distinct Schema.org type count per page (mean 7.0, median 7, max 29). No pages with exactly 6 types appear in the data; the 12+ bar groups all remaining pages.

### 4.7 Annotation Richness: Average Properties per Entity (AvP)

The *Average Properties per Entity* (AvP) measures the mean number of distinct properties populated in each entity instance. Formally, for a set of entities $E$:

$$\text{AvP} = \frac{1}{|E|} \sum_{e \in E} |P(e)| \tag{1}$$

where $P(e)$ is the set of properties associated with entity $e$. Higher AvP indicates richer, more detailed structured annotations, while lower values reflect minimal or sparse markup. AvP can be used as a proxy for annotation depth in `schema.org` usage across web pages.

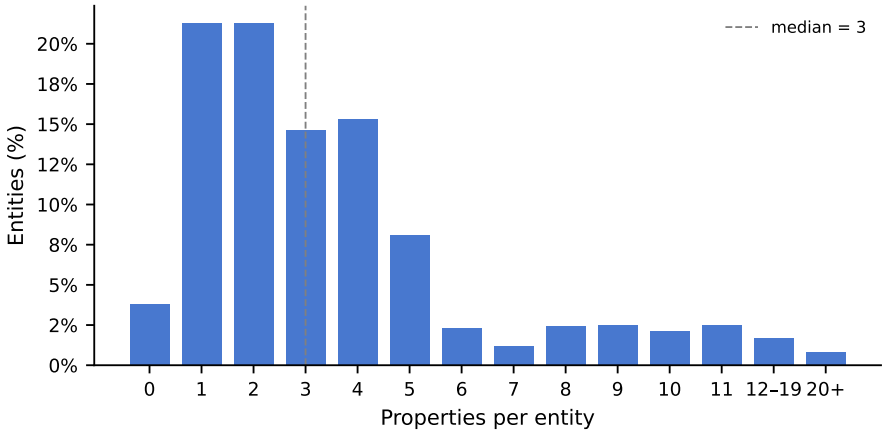



Figure 3: Corpus-wide AvP distribution (mean 3.7, median 3.0, SD 3.6). The dashed line marks the median.

Over 57% of entities have 3 or fewer properties populated, and the median of 3.0 confirms that minimal annotation is the modal behaviour across the corpus. The long tail above 10 properties (under 8% of entities) represents the richest annotations in the dataset, but even these cover only a small fraction of each type's defined vocabulary.

Table 11: AvP breakdown by Schema.org type for example types (full results for all 47 analysed types in Appendix).

| Type | Entities | Avail. props | AvP mean | Median |
|---|---|---|---|---|
| Article | 172,671 | 136 | 19.57 | 25.0 |
| NewsArticle | 199,809 | 141 | 12.61 | 14.0 |
| ClaimReview | 8,729 | 139 | 6.60 | 6.0 |
| Event | 79,947 | 56 | 6.58 | 5.0 |
| Organization | 1,009,766 | 89 | 5.92 | 4.0 |
| Rating | 11,486 | 19 | 2.88 | 3.0 |
| Product | 45,456 | 72 | 2.81 | 2.0 |
| Person | 410,829 | 81 | 2.79 | 2.0 |
| Claim | 4,007 | 132 | 2.62 | 3.0 |
| Place | 580,546 | 60 | 2.02 | 2.0 |

`Article` entities are the most richly annotated content type (AvP 19.57), yet even they use fewer than 15% of their 136 defined properties on average.

### 4.8 Property Coverage per Type

For each type, *property coverage* is the average fraction of schema.org-defined properties used per entity (entities using no defined property contribute zero to the average). Across all 137 types in the corpus, the macro-averaged coverage (i.e., the unweighted mean of per-type coverage values,

where each type contributes equally regardless of its number of entities) is 5.75% ($\sigma = 4.64\%$), reflecting that most entities instantiate only a small subset of their type's defined vocabulary.

Table 12: Average per-entity property coverage for example types (full results for all 47 analysed types in Appendix).

| **Type** | **Entities** | **Defined** | **Avg Coverage** |
|---|---|---|---|
| Rating | 11,516 | 19 | 15.09% |
| Article | 172,671 | 136 | 14.38% |
| Event | 80,561 | 56 | 11.63% |
| NewsArticle | 199,809 | 141 | 8.79% |
| Organization | 1,010,110 | 89 | 6.64% |
| ClaimReview | 8,729 | 139 | 4.74% |
| Product | 45,457 | 72 | 3.88% |
| Person | 410,961 | 81 | 3.44% |
| Place | 580,598 | 60 | 3.36% |
| Claim | 4,007 | 132 | 1.98% |
| *Macro avg (137 types)* | | | 5.75% |
| *Std dev* | | | 4.64% |

### 4.9 Characteristic Property Sets

A *characteristic set* is an exact set of schema.org properties used by an entity. Tables below show the three most frequent sets for each of the ten example types; full results for all 47 analysed types are provided in the Appendix.

**ClaimReview (24 unique sets; 139 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 4,107 (47.1%) | {author, claimReviewed, datePublished, itemReviewed, reviewRating, url} |
| 2 | 1,719 (19.7%) | {author, claimReviewed, datePublished, description, itemReviewed, reviewRating, url} |
| 3 | 775 (8.9%) | {author, claimReviewed, datePublished, reviewRating, url} |

**Claim (13 unique sets; 132 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 2,533 (63.2%) | {appearance, author, datePublished} |
| 2 | 520 (13.0%) | {datePublished} |
| 3 | 410 (10.2%) | {author, datePublished, url} |

**NewsArticle (224 unique sets; 141 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 41,330 (20.7%) | {author, dateModified, datePublished, headline, image, mainEntityOfPage, publisher} |
| 2 | 18,582 (9.3%) | {articleBody, articleSection, author, dateModified, datePublished, description, hasPart, headline, image, inLanguage, isAccessibleForFree, isPartOf, keywords, mainEntityOfPage, publisher, url} |
| 3 | 17,602 (8.8%) | {articleBody, articleSection, author, dateCreated, dateModified, datePublished, description, headline, image, keywords, locationCreated, mainEntityOfPage, publisher, publishingPrinciples} |

**Article (133 unique sets; 136 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 49,185 (28.5%) | {author, dateModified, datePublished, headline, image, mainEntity, name, publisher, sameAs, url} |
| 2 | 17,949 (10.4%) | {author, dateModified, datePublished, headline, mainEntity, name, publisher, sameAs, url} |
| 3 | 16,179 (9.4%) | {alternativeHeadline, associatedMedia, author, copyrightHolder, copyrightYear, creator, dateCreated, dateModified, datePublished, description, discussionUrl, genre, headline, image, inLanguage, keywords, license, mainEntityOfPage, name, publisher, publishingPrinciples, sourceOrganization, speakable, thumbnailUrl, url} |

**Organization (63 unique sets; 89 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 295,728 (29.3%) | {address, aggregateRating, award, email, image, legalName, logo, name, sameAs, telephone, url} |
| 2 | 215,329 (21.3%) | {name} |
| 3 | 166,687 (16.5%) | {logo, name} |

**Person (170 unique sets; 81 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 128,766 (31.3%) | {name, url} |
| 2 | 124,946 (30.4%) | {name} |
| 3 | 55,459 (13.5%) | {image, name, url} |

**Event (52 unique sets; 56 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 47,077 (58.4%) | {endDate, location, name, startDate, url} |
| 2 | 29,364 (36.4%) | {description, endDate, eventStatus, image, location, name, performer, startDate, url} |
| 3 | 614 (0.8%) | {(no SO properties)} |

**Place (18 unique sets; 60 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 519,513 (89.5%) | {address, name} |
| 2 | 28,241 (4.9%) | {name} |
| 3 | 14,241 (2.5%) | {address, geo, hasMap, name, url} |

**Product (31 unique sets; 72 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 38,792 (85.3%) | {name, productID} |
| 2 | 3,048 (6.7%) | {brand, category, description, image, name, offers, sku, url} |
| 3 | 849 (1.9%) | {brand, category, description, image, name, offers, releaseDate, sku, url} |

**Rating (9 unique sets; 19 available properties).**

| # | Count (%) | Property set |
|---|---|---|
| 1 | 4,104 (35.6%) | {alternateName, bestRating, ratingValue, worstRating} |
| 2 | 3,603 (31.3%) | {alternateName} |
| 3 | 2,593 (22.5%) | {bestRating, ratingValue, worstRating} |

Overall, characteristic property sets are highly concentrated, with a small number of repeated annotation patterns accounting for most entities of each type. This suggests that structured data on the Web is usually generated from reusable templates (predefined layouts that automatically fill in metadata fields), rather than being carefully tailored to each individual entity. Although richer property combinations exist, they are uncommon and contribute only marginally to overall coverage.

### 4.10 Property Fill Rate by Domain

Fill rate per entity = distinct properties used / properties defined for the entity's type. Corpus-wide mean fill rate: **7.4%** (11,823,497 typed entities; 45,124,082 distinct entity–predicate pairs). Domain-level aggregation is at the registered-domain level (e.g. subdomains of `cnn.com` are merged).

Table 13: Property fill rate for selected fact-checking organisations, news outlets, and knowledge bases, aggregated at the registered-domain level, sorted by fill rate within each group (descending). *Entities*: typed entities from this domain; *Avail.*: avg. number of Schema.org properties applicable to the entity's declared type; *Avg Used*: mean properties used per entity; *Fill%*: avg used / avail.

| **Domain** | **Entities** | **Avail.** | **Avg Used** | **Fill%** |
|---|---|---|---|---|
| *Fact-checking organisations* | | | | |
| politifact.com | 3,935 | 91 | 4.10 | 8.0% |
| afp.com | 2,373 | 93 | 3.61 | 6.6% |
| vishvasnews.com | 16,987 | 85 | 3.57 | 5.8% |
| facta.news | 8,631 | 70 | 3.12 | 5.8% |
| demagog.org.pl | 23,075 | 75 | 3.05 | 5.8% |
| maldita.es | 6,094 | 107 | 3.13 | 4.8% |
| factcheck.org | 17,539 | 96 | 2.70 | 4.8% |
| leadstories.com | 17,200 | 108 | 3.40 | 4.5% |
| fullfact.org | 9,442 | 97 | 2.86 | 3.2% |
| *News outlets (global)* | | | | |
| cbsnews.com | 249,990 | 99 | 3.80 | 5.6% |
| cnn.com | 281,214 | 113 | 5.02 | 4.3% |
| aljazeera.com | 2,971 | 117 | 2.97 | 2.9% |
| dw.com | 208,338 | 121 | 2.74 | 2.3% |
| *News outlets (national/regional)* | | | | |
| lefigaro.fr | 339,409 | 45 | 3.20 | 10.7% |
| ria.ru | 3,537,108 | 78 | 5.08 | 10.0% |
| elpais.com | 1,148,493 | 53 | 2.85 | 9.6% |
| rbc.ua | 408,030 | 62 | 4.20 | 9.3% |
| elcomercio.pe | 32,489 | 114 | 6.94 | 6.6% |
| lequipe.fr | 11,259 | 83 | 3.63 | 6.6% |
| lavanguardia.com | 18,085 | 119 | 7.24 | 6.1% |
| clarin.com | 5,193 | 86 | 3.70 | 5.9% |
| indiatimes.com | 499,882 | 104 | 3.39 | 4.5% |
| freepressjournal.in | 659,227 | 109 | 2.88 | 4.3% |
| pravda.ru | 57,486 | 36 | 1.51 | 4.2% |
| infobae.com | 70,679 | 128 | 3.19 | 2.6% |
| *Knowledge base* | | | | |
| wikipedia.org | 277,768 | 116 | 3.42 | 2.8% |

Fact-checking organisations cluster in the 3.2–8.0% fill range, while news outlets and knowledge bases show similarly low values, indicating a consistent pattern of underutilization of schema.org properties across domains. Even the highest-performing cases rarely exceed 10% fill.

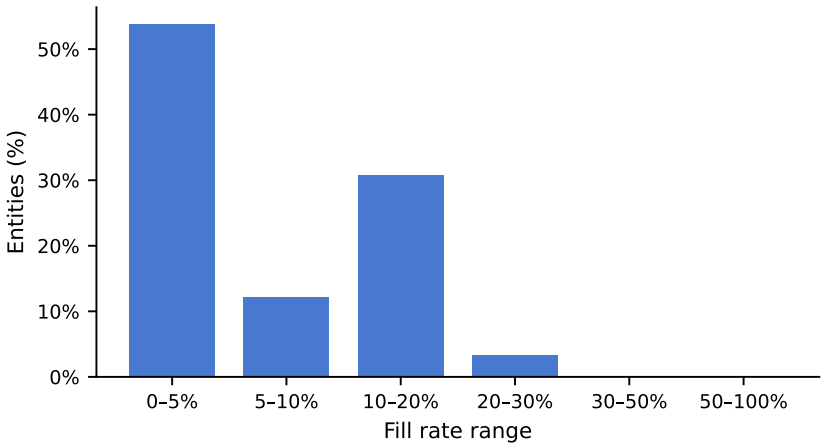



Figure 4: Distribution of per-entity fill rates across the full typed-entity corpus (11,823,497 entities). The 30–50% and 50–100% bins each contain fewer than 20 entities (<0.1%) and are not visible at this scale.

More than half (53.7%) of all typed entities fall below 5% fill rate, and essentially no entities exceed 30%. This corpus-wide sparsity quantifies the gap that LLM-based extraction is intended to close: existing annotations provide a syntactically correct but semantically sparse representation, leaving most of each type's vocabulary unpopulated.

### 4.11 Article and NewsArticle Pages

News and article pages are primary candidate sources for automated fact-checking: they carry the claims, contextual evidence, and source references that fact-checking pipelines need to reason over. To understand how much structured semantic information Schema.org annotations already provide on such pages, we identify a representative set of article and news-article pages from the corpus and analyse the richness and coverage of their existing annotations.

Ideally, annotation completeness would be assessed by running NLP pipelines (named-entity recognition, relation extraction) over the raw text of each page to identify what entities and relationships the content actually contains, then comparing that picture against the existing Schema.org markup to measure the gap. At the scale of hundreds of thousands of pages this is computationally expensive, and having sample would also raise the question of how representative it is of the article corpus. So for now, this is left for future work.

As a first step, we instead examine what annotations are already present: how richly they describe article content, which properties are populated, and which Schema.org types one would expect from news and factual reporting are absent or severely underrepresented.

Pages were identified using four complementary signals: single entity (Article and/or NewsArticle) per page, subject URI matching the page URI, presence of `headline` + `datePublished`/author/body, and URL path heuristics.

Table 14: Article page identification.

| Filter | Pages |
|---|---|
| Candidate pages | 356,132 |
| Signal 1: single Article/NewsArticle entity per page | 342,808 |
| Signal 2: subject URL matches page URL | 5,613 |
| Signal 3: has headline + date/author/body | 354,084 |
| Signal 4: URL path looks like article | 354,740 |
| Strict intersection (all 4 signals) | 5,610 |
| **Lenient intersection (signals 1+3)** | **340,789** |

The lenient set (signals 1+3) is adopted as the article-page corpus used for the analysis in this section.

We examine four Schema.org properties with direct semantic relevance to fact-checking: `articleBody` (full article text), `about` (the entity the article reports on), `citation` (external sources cited by the author, supporting source verification), and `mentions` (named entities referenced in the text, which if populated would provide structured named-entity annotations without requiring additional extraction from raw text). Table 15 reports their actual fill rates.

Table 15: Content property fill rates on article pages.

| Property | Article | NewsArticle | Combined | Note |
|---|---|---|---|---|
| `articleBody` | 32.2% | 61.6% | 47.8% | full text |
| `about` | 39.6% | 1.3% | 19.3% | topic entity |
| `citation` | 23.3% | 0.3% | 11.1% | references |
| `mentions` | <0.1% | <0.1% | <0.1% | named entities |

*NewsArticle detail (178,854 entities):* `articleBody` 110,241 (61.6%); `about` 2,286 (1.3%); `citation` 448 (0.3%); `mentions` 27 (<0.1%).

**Structure of `about` targets.** Of the 65,146 Article and NewsArticle entities with an `about` link (19.3%), the linked targets comprise 63,411 distinct entities across 9 types. Among those entities:

- `Thing`: 62,955 (99.3%): bare entities with no further structured properties.
- `SportsTeam`: 190 (0.3%)
- `SportsEvent`: 180 (0.3%)
- `Event`: 58 (0.1%)
- `GovernmentOrganization`: 12; `Person`: 8; `EducationalOrganization`: 4; `Country`: 2; `Organization`: 2

Sample `name` values from `Thing` targets reveal that these are not structured entities but free-text topic strings, typically comma-separated keyword lists, providing no usable structure.

**Characteristic property sets.** To understand *what* is annotated on these pages and how richly, we examine the characteristic property sets: distinct combinations of Schema.org properties assigned to entities. Table 17 shows the top-3 most common sets for all 23 entity types on article pages: `Article`, `NewsArticle`, and the 21 co-occurring types catalogued in Table 16.

Table 16: Co-occurring entity types on Article/NewsArticle pages (base: 340,789 lenient article pages).

| Type | Pages | % of article pages |
|---|---|---|
| ImageObject | 340,654 | 100.0% |
| Organization | 297,085 | 87.2% |
| WebPage | 204,916 | 60.1% |
| NewsArticle | 182,175 | 53.5% |
| PostalAddress | 156,314 | 45.9% |
| NewsMediaOrganization | 156,825 | 46.0% |
| Article | 158,695 | 46.6% |
| ListItem | 168,463 | 49.4% |
| BreadcrumbList | 168,208 | 49.4% |
| Person | 137,195 | 40.3% |
| WebSite | 112,988 | 33.2% |
| AggregateRating | 96,115 | 28.2% |
| Country | 95,034 | 27.9% |
| Thing | 78,889 | 23.1% |
| SpeakableSpecification | 76,483 | 22.4% |
| CreativeWork | 50,186 | 14.7% |
| AdministrativeArea | 48,074 | 14.1% |
| ContactPoint | 44,991 | 13.2% |
| WebPageElement | 34,852 | 10.2% |
| Product | 32,403 | 9.5% |
| Place | 27,458 | 8.1% |
| VideoObject | 22,211 | 6.5% |
| CollectionPage | 12,836 | 3.8% |

The near-universal presence of `ImageObject` (100%) and `Organization` (87.2%) mirrors the standard CMS template: every article embeds images and a publisher entity regardless of content. Types directly representing fact-checks (`ClaimReview`, `Claim`) are entirely absent from these article pages, confirming that fact-check annotations exist only on dedicated fact-check pages and are not co-annotated on general news articles.

Table 17: Top-3 characteristic property sets for all entity types on article pages (340,789 lenient article pages). % = fraction of entities of that type within the article-page corpus.

| # | Entities | % | Properties |
|---|---|---|---|
| *Article* (158,695 entities; 136 available properties) | | | |
| 1 | 45,131 | 28.4 | author, dateModified, datePublished, headline, image, mainEntity, name, publisher, sameAs, url |
| 2 | 17,393 | 11.0 | author, dateModified, datePublished, headline, mainEntity, name, publisher, sameAs, url |
| 3 | 15,905 | 10.0 | alternativeHeadline, associatedMedia, author, copyrightHolder, copyrightYear, creator, dateCreated, dateModified, datePublished, description, discussionUrl, genre, headline, image, inLanguage, keywords, license, mainEntityOfPage, name, publisher, publishingPrinciples, sourceOrganization, speakable, thumbnailUrl, url |
| *NewsArticle* (178,854 entities; 141 available properties) | | | |
| 1 | 40,755 | 22.8 | author, dateModified, datePublished, headline, image, mainEntityOfPage, publisher |

*continued. . .*

Table 17 (continued)

| # | Entities | % | Properties |
|---|---|---|---|
| 2 | 18,450 | 10.3 | articleBody, articleSection, author, dateModified, datePublished, description, hasPart, headline, image, inLanguage, isAccessibleForFree, isPartOf, keywords, mainEntityOfPage, publisher, url |
| 3 | 17,548 | 9.8 | articleBody, articleSection, author, dateCreated, dateModified, datePublished, description, headline, image, keywords, locationCreated, mainEntityOfPage, publisher, publishingPrinciples |
| *ImageObject* (1,561,640 entities; 151 available properties) | | | |
| 1 | 767,153 | 49.1 | contentUrl, height, url, width |
| 2 | 392,827 | 25.2 | height, url, width |
| 3 | 189,519 | 12.1 | contentUrl, height, representativeOfPage, url, width |
| *Organization* (859,726 entities; 89 available properties) | | | |
| 1 | 294,265 | 34.2 | address, aggregateRating, award, email, image, legalName, logo, name, sameAs, telephone, url |
| 2 | 162,374 | 18.9 | name |
| 3 | 158,699 | 18.5 | address, aggregateRating, award, email, legalName, logo, name, sameAs, telephone, url |
| *ListItem* (568,698 entities; 17 available properties) | | | |
| 1 | 284,856 | 50.1 | item, name, position, url |
| 2 | 147,397 | 25.9 | item, position |
| 3 | 130,009 | 22.9 | item, name, position |
| *PostalAddress* (540,246 entities; 30 available properties) | | | |
| 1 | 470,697 | 87.1 | addressCountry, addressLocality, addressRegion, postalCode, streetAddress |
| 2 | 32,460 | 6.0 | addressCountry, addressLocality, postalCode, streetAddress |
| 3 | 17,007 | 3.1 | *(no properties)* |
| *AggregateRating* (454,443 entities; 22 available properties) | | | |
| 1 | 453,015 | 99.7 | bestRating, ratingCount, ratingValue, worstRating |
| 2 | 1,369 | 0.3 | bestRating, ratingCount, ratingValue |
| 3 | 45 | <0.1 | ratingCount, ratingValue |
| *Country* (453,044 entities; 60 available properties) | | | |
| 1 | 453,039 | 100.0 | name |
| 2 | 2 | <0.1 | description, name, url |
| 3 | 1 | <0.1 | item, name, position |
| *Person* (210,433 entities; 81 available properties) | | | |
| 1 | 81,050 | 38.5 | name, url |
| 2 | 75,654 | 36.0 | name |
| 3 | 16,444 | 7.8 | image, jobTitle, name, sameAs, url |
| *WebPage* (204,916 entities; 139 available properties) | | | |
| 1 | 88,240 | 43.1 | *(no properties)* |
| 2 | 31,043 | 15.1 | name |
| 3 | 27,487 | 13.4 | name, speakable, url |
| *BreadcrumbList* (170,358 entities; 17 available properties) | | | |

*continued...*

Table 17 (continued)

| # | Entities | % | Properties |
|---|---|---|---|
| 1 | 169,694 | 99.6 | itemListElement |
| 2 | 664 | 0.4 | itemListElement, name |
| *NewsMediaOrganization* (156,825 entities; 93 available properties) | | | |
| 1 | 53,871 | 34.4 | logo, name |
| 2 | 37,564 | 24.0 | logo, name, url |
| 3 | 30,114 | 19.2 | foundingDate, logo, name, sameAs, url |
| *WebSite* (102,011 entities; 130 available properties) | | | |
| 1 | 94,955 | 93.1 | description, image, name, url |
| 2 | 3,843 | 3.8 | potentialAction, url |
| 3 | 3,179 | 3.1 | name, url |
| *Thing* (79,127 entities; 13 available properties) | | | |
| 1 | 75,242 | 95.1 | name |
| 2 | 3,756 | 4.7 | name, url |
| 3 | 124 | 0.2 | name, sameAs |
| *SpeakableSpecification* (76,557 entities; 15 available properties) | | | |
| 1 | 44,072 | 57.6 | cssSelector |
| 2 | 32,424 | 42.4 | xpath |
| 3 | 46 | 0.1 | cssSelector, xPath |
| *AdministrativeArea* (57,499 entities; 60 available properties) | | | |
| 1 | 57,493 | 100.0 | name |
| 2 | 6 | <0.1 | description, name, url |
| *ContactPoint* (54,341 entities; 23 available properties) | | | |
| 1 | 29,616 | 54.5 | areaServed, availableLanguage, contactType, email, telephone |
| 2 | 17,366 | 32.0 | contactType, telephone |
| 3 | 6,522 | 12.0 | contactType, email |
| *CreativeWork* (51,331 entities; 129 available properties) | | | |
| 1 | 31,539 | 61.4 | name, productID |
| 2 | 16,019 | 31.2 | name, url |
| 3 | 1,707 | 3.3 | author, datePublished, keywords |
| *WebPageElement* (34,852 entities; 131 available properties) | | | |
| 1 | 34,852 | 100.0 | cssSelector, isAccessibleForFree |
| *Product* (32,627 entities; 72 available properties) | | | |
| 1 | 31,539 | 96.7 | name, productID |
| 2 | 551 | 1.7 | Name, brand, description, image, mpn, productID, sku |
| 3 | 241 | 0.7 | name |
| *Place* (36,385 entities; 60 available properties) | | | |
| 1 | 18,035 | 49.6 | address, name |
| 2 | 14,537 | 40.0 | name |

*continued...*

Table 17 (continued)

| # | Entities | % | Properties |
|---|---|---|---|
| 3 | 3,586 | 9.9 | address |
| *VideoObject* (28,631 entities; 157 available properties) | | | |
| 1 | 8,723 | 30.5 | contentUrl, dateModified, description, duration, embedUrl, headline, inLanguage, mainEntityOfPage, name, thumbnail, thumbnailUrl, uploadDate |
| 2 | 7,323 | 25.6 | contentUrl, description, duration, embedUrl, mainEntityOfPage, name, publisher, thumbnail, thumbnailUrl, uploadDate |
| 3 | 4,079 | 14.2 | caption, contentUrl, dateModified, description, duration, embedUrl, headline, inLanguage, mainEntityOfPage, name, thumbnail, thumbnailUrl, uploadDate |
| *CollectionPage* (994 entities; 139 available properties) | | | |
| 1 | 672 | 67.6 | description, hasPart, headline, url |
| 2 | 257 | 25.9 | `@name` (invalid property) |
| 3 | 65 | 6.5 | name |

Across all 23 types, annotations appear to be shaped by templates rather than the content of individual pages. `Article` entities are the richest (AvP 20.1), built around a consistent metadata core (`author`, `datePublished`, `headline`, `image`, `publisher`, `url`); `NewsArticle` is more compact (AvP 12.6), with the most common set covering just seven properties and richer variants adding access restrictions (`isAccessibleForFree`, `isPartOf`) and location metadata.

The co-occurring types follow the same pattern. `BreadcrumbList` and `AggregateRating` are effectively single-property types; `Country` and `AdministrativeArea` converge at over 99% on `name` alone; `ImageObject` carries only technical dimensions with no caption or authorship; `WebPage` is the sparsest, with 43.1% of instances carrying no properties at all; and `Person` entities are thin, with 74.5% reduced to `name` or `name`, `url`.

Content properties are almost entirely absent: `mentions` fills at under 0.1%, `citation` at 11.1%, and `about` at 19.3%, with 99.3% of its targets being bare `Thing` nodes carrying only a name string.

Beyond individual properties, entire Schema.org types that would naturally describe the content of news articles are missing from the co-occurring entity inventory. News articles report on events, actions, and evolving situations, yet `Event` appears in only 58 `about` targets (0.1%), making it effectively absent from article pages. Types covering actions and decisions, including `LegislativeAction`, `GovernmentAction`, `VoteAction`, `CommunicateAction`, and related `Action` subtypes, are entirely absent, even though these are the vocabulary one would reach for to describe decisions, political developments, and public statements. `Legislation` and types encoding factual claims, such as `Claim` and `Statement`, are likewise absent. Geographic entities are present but shallow: `Country` (27.9%) and `AdministrativeArea` (14.1%) both converge at 99–100% on `name` alone.

Structured data on article pages therefore describes publication infrastructure rather than semantic content, motivating the use of LLMs to extract the richer, content-grounded annotations that Schema.org publishers currently omit.

# 5 ClaimReview Analysis

This section reports on the full WDC 2024-12 ClaimReview class-specific subset (not restricted to the target domain list), enabling a broad assessment of how fact-checking organisations use Schema.org [3].

## 5.1 Dataset

Unlike the domain-filtered corpus in the preceding sections, this dataset covers all `ClaimReview` entities across all crawled domains, providing the most comprehensive available snapshot of how fact-checking organisations deploy this vocabulary worldwide.

Table 18: ClaimReview dataset overview.

| **Metric** | **Value** |
|---|---|
| Source file (`part_0.gz`) | 57.6 MB compressed |
| N-Quad lines parsed | 3,919,703 |
| All entities parsed | 787,354 |
| **ClaimReview entities** | **60,313** |
| Registered domains | 318 |
| *Schema.org vocabulary for ClaimReview:* | |
| Properties defined for ClaimReview | 139 |
| Properties actually used | 41 (29.5%) |
| Properties never used | 98 (70.5%) |

## 5.2 Syntactic Validity

The same four syntactic validity checks applied to the full corpus in Section 4 are repeated here for `ClaimReview` specifically, establishing whether the class-specific data is cleaner or noisier than the corpus average.

Table 19: ClaimReview syntactic quality metrics.

| **Metric** | **Value** | **%** |
|---|---|---|
| Namespace correctness | 316,261 / 316,261 | 100.00% |
| Type validity | 60,539 / 60,540 | 100.00% |
| Property validity | 255,057 / 255,721 | 99.74% |
| Range violations | 5,858 / 258,093 | 2.27% |

ClaimReview is significantly cleaner than the full-corpus average on all four metrics. The single type invalidity is `ExplainerNewsArticle` (a non-standard extension type). Range violations are dominated by non-ISO date formats in `datePublished` (4,990 cases; e.g., `"10/08/2023"`, `"3 Nov. 2018"`) and strings in `potentialAction` where an `Action` node is expected (488 cases).

## 5.3 Property Coverage

Property coverage here identifies which ClaimReview vocabulary fields are actively used and at what density, defined as the fraction of ClaimReview entities that populate each property. A low density for core semantic fields (`claimReviewed`, `reviewRating`, `itemReviewed`) directly quantifies how much of the fact-check intended semantics is unfulfilled in practice.

Table 20: Top 20 used ClaimReview properties, sorted by entity density (density = fraction of ClaimReview entities that populate the property).

| Property | Entities using | Density % |
|---|---|---|
| url | 36,466 | 60.46% |
| author | 36,274 | 60.14% |
| claimReviewed | 36,020 | 59.72% |
| reviewRating | 35,766 | 59.30% |
| itemReviewed | 33,730 | 55.92% |
| datePublished | 32,992 | 54.70% |
| dateModified | 6,813 | 11.30% |
| image | 6,306 | 10.46% |
| publisher | 5,649 | 9.37% |
| mainEntityOfPage | 5,278 | 8.75% |
| name | 5,049 | 8.37% |
| description | 4,949 | 8.21% |
| keywords | 1,366 | 2.26% |
| inLanguage | 1,345 | 2.23% |
| headline | 1,291 | 2.14% |
| reviewBody | 1,239 | 2.05% |
| sdPublisher | 879 | 1.46% |
| isPartOf | 735 | 1.22% |
| potentialAction | 718 | 1.19% |
| dateCreated | 385 | 0.64% |

Approximately 40% of ClaimReview entities lack `claimReviewed`, `reviewRating`, `itemReviewed`. The noisy `vpravda.ru` cluster (Section 5.7) accounts for most of this gap.

### 5.4 Annotation Richness (AvP)

AvP for ClaimReview is computed identically to Section 4.7, but interpreted against the class's minimal semantic contract: a usable `ClaimReview` record requires at minimum a claim text (`claimReviewed`), a verdict (`reviewRating`), and a link to the reviewed item (`itemReviewed`).

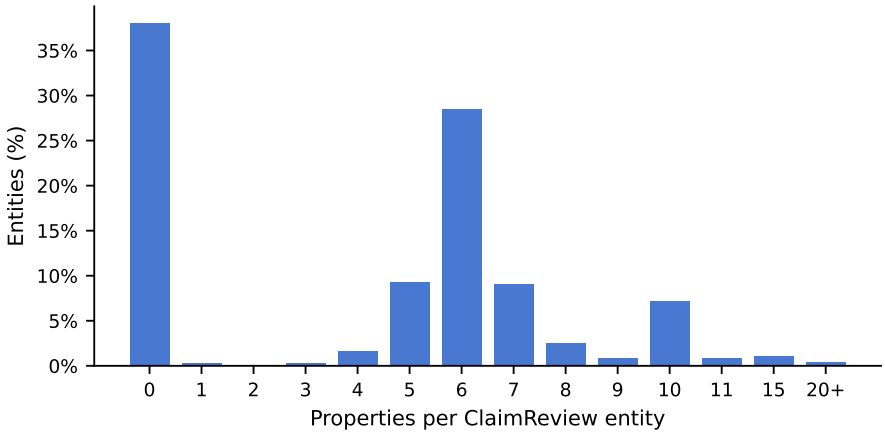



Figure 5: ClaimReview AvP distribution (mean 4.24, median 6.0, SD 3.79, max 24). The shape reflects the zero-property vpravda.ru noise cluster (38%) and the 6-property fact-check structure (28.5%).

The bimodal shape, with 38% of entities at 0 properties and the largest non-zero bin at 6 properties (28.5%), maps directly onto the vpravda.ru noise cluster versus the typical fact-checking structure used by established organisations (the 6-property set: author, claimReviewed, datePublished, itemReviewed, reviewRating, url). Excluding zero-property entities, the effective mean AvP rises to approximately 6.8, corresponding to about 4.9% of the 139 defined properties, still well below the semantic richness the vocabulary supports.

### 5.5 Characteristic Property Sets

Characteristic property sets reveal which combinations of semantic fields publishers use together and their relative frequency.

Table 21: Top 15 ClaimReview characteristic property sets (105 unique sets total).

| # | n (%) | Property set |
|---|---|---|
| 1 | 22,896 (38.0%) | {none, only `rdf:type`} |
| 2 | 16,737 (27.8%) | {author, claimReviewed, datePublished, itemReviewed, reviewRating, url} |
| 3 | 3,885 (6.4%) | {author, claimReviewed, dateModified, datePublished, image, itemReviewed, mainEntityOfPage, publisher, reviewRating, url} |
| 4 | 3,335 (5.5%) | {author, claimReviewed, itemReviewed, reviewRating, url} |
| 5 | 2,329 (3.9%) | {author, claimReviewed, datePublished, itemReviewed, name, reviewRating, url} |
| 6 | 1,719 (2.9%) | {author, claimReviewed, datePublished, description, itemReviewed, reviewRating, url} |
| 7 | 1,091 (1.8%) | {author, claimReviewed, datePublished, reviewRating, url} |
| 8 | 963 (1.6%) | {author, claimReviewed, datePublished, description, itemReviewed, name, reviewRating, url} |
| 9 | 878 (1.5%) | {author, claimReviewed, datePublished, itemReviewed, reviewRating, sdPublisher, url} |
| 10 | 626 (1.0%) | {author, claimReviewed, dateModified, datePublished, description, headline, image, inLanguage, itemReviewed, keywords, mainEntityOfPage, name, publisher, reviewRating, url} |
| 11 | 488 (0.8%) | {description, image, name, potentialAction, url} |
| 12 | 427 (0.7%) | {author, claimReviewed, dateModified, datePublished, description, itemReviewed, reviewBody, reviewRating, url} |
| 13 | 386 (0.6%) | {author, claimReviewed, dateModified, datePublished, description, itemReviewed, keywords, reviewBody, reviewRating, url} |
| 14 | 368 (0.6%) | {author, claimReviewed, dateCreated, dateModified, datePublished, itemReviewed, reviewRating, url} |
| 15 | 340 (0.6%) | {author, claimReviewed, datePublished, image, inLanguage, isPartOf, itemReviewed, mainEntityOfPage, publisher, reviewRating, url} |

Set #1 (38%) is entirely the `vpravda.ru` cluster: entities carry only a `ClaimReview` type

declaration with no semantic properties and are effectively noise. The dominant legitimate set (#2, 27.8%) contains the six core semantic fields. The remaining sets vary primarily in optional metadata fields (image, description, keywords).

### 5.6 Structure of `itemReviewed`

The `itemReviewed` property links a `ClaimReview` to the item being reviewed, and its nested `appearance` property is the mechanism for pointing to where the claim originally circulated (a social media post, news article, or broadcast). The range of `itemReviewed` is `Thing`, so any Schema.org type is valid.

Corpus-wide, only 55.9% of ClaimReview entities populate `itemReviewed`, but this is dominated by the `vpravda.ru` noise cluster (38% of all entities, zero properties). Among the 37,422 non-noise entities, approximately 90% link to a reviewed item.

Table 22: Types of entities linked via `itemReviewed` (33,773 entities).

| **Type** | **Count** | **%** |
|---|---|---|
| Claim | 19,824 | 58.7% |
| CreativeWork | 13,930 | 41.2% |
| creativeWork (invalid capitalisation) | 17 | 0.1% |
| NewsArticle | 2 | 0.0% |

Of the 33,773 linked entities, only sets 1 and 7 in Table 23 include `appearance`, covering 40.2% of targets. The remaining 59.8% record only provenance metadata (`author`, `datePublished`) with no source link.

Table 23: Top 10 characteristic property sets for `itemReviewed` targets.

| # | n (%) | Property set |
|---|---|---|
| 1 | 12,832 (38.0%) | {appearance, author, datePublished} |
| 2 | 7,156 (21.2%) | {author, datePublished} |
| 3 | 3,885 (11.5%) | {author, dateModified, datePublished, name} |
| 4 | 2,532 (7.5%) | {author, datePublished, url} |
| 5 | 1,719 (5.1%) | {author, datePublished, keywords} |
| 6 | 1,007 (3.0%) | {author, datePublished, image, name} |
| 7 | 736 (2.2%) | {appearance, author, datePublished, firstAppearance} |
| 8 | 647 (1.9%) | {author} |
| 9 | 644 (1.9%) | {datePublished} |
| 10 | 618 (1.8%) | {author, dateModified, datePublished, image, mainEntityOfPage, name} |

Resolving the 23,943 `appearance` links reveals that 81.7% point to an external domain (a social media post, third-party article, or broadcast source), confirming that when `appearance` is used it functions as intended. The remaining 14.9% are self-referential, pointing back to the publisher's own site; inspection shows this pattern is concentrated in a small number of publishers, including spam-like domains and one legitimate outlet (`tempo.co`) that links to its own fact-check pages rather than the original claim source. Among established fact-checking organisations (e.g. `fullfact.org`), external linking is consistent with no self-referential cases.

The limiting factor is therefore not misuse but coverage: of non-noise ClaimReview entities, approximately 90% link to an `itemReviewed` target, and only 40% of those include `appearance`,

yielding end-to-end coverage of roughly 36% for a structured source link across the legitimate fact-checking corpus.

### 5.7 Domain-Level Annotation Quality

Domain-level fill rates expose the gap between publishers that implement the ClaimReview schema thoroughly and those that contribute structurally empty or misapplied entries, and identify which organisations produce data reliable enough to serve as evaluation baselines for LLM extraction.

Table 24: Domain-level ClaimReview annotation quality (top 20 by entity count). **%**: share of all 60,313 ClaimReview entities; **fill%**: avg fraction of the 139 defined ClaimReview properties populated per entity (overall avg: 3.1%).

| Domain | n | % | fill% |
|---|---|---|---|
| vpravda.ru | 22,891 | 38.0% | 0.0% |
| afp.com | 2,918 | 4.8% | 4.7% |
| co.th | 2,337 | 3.9% | 4.3% |
| boomlive.in | 2,152 | 3.6% | 7.2% |
| leadstories.com | 1,720 | 2.9% | 5.0% |
| org.pl | 1,667 | 2.8% | 4.3% |
| healthday.com | 1,330 | 2.2% | 4.3% |
| provereno.media | 1,255 | 2.1% | 4.3% |
| types-cars.com | 1,157 | 1.9% | 4.3% |
| vishvasnews.com | 1,000 | 1.7% | 4.3% |
| misbar.com | 963 | 1.6% | 5.8% |
| com.br | 894 | 1.5% | 4.7% |
| fullfact.org | 855 | 1.4% | 6.8% |
| politifact.com | 787 | 1.3% | 4.3% |
| maldita.es | 769 | 1.3% | 3.6% |
| tempo.co | 629 | 1.0% | 3.6% |
| truthorfiction.com | 609 | 1.0% | 3.6% |
| correctiv.org | 577 | 1.0% | 4.3% |
| newsmeter.in | 567 | 0.9% | 7.2% |
| facta.news | 567 | 0.9% | 4.3% |

The corpus-wide average fill rate is **3.1%** across all 60,313 ClaimReview entities, driven almost downward by **vpravda.ru** (Volgograd regional newspaper, Russia), which contributes 38% of all entities with a fill rate of 0%. Excluding this cluster, established fact-checking organisations reach fill rates of 3.6–7.2%, with `boomlive.in` (7.2%) and `fullfact.org` (6.8%) annotating most richly relative to the other domains.

### 5.8 Example Entities

**Depth-2 entity (typical ClaimReview structure).**

```
types:  ClaimReview
graph:  https://pagellapolitica.it/fact-checking/...
author → [Organization]{name="Pagella Politica", url=...}
claimReviewed = "Ecco i costi della regolarizzazione..."
itemReviewed → [Claim]{appearance=..., author=→[Person], datePublished="2020-06-4"}
reviewRating → [Rating]{alternateName="Ni'"}
url = <https://pagellapolitica.it/...>
```

**Flat entity (depth 0, vpravda.ru noise).**

types: ClaimReview
graph: https://vpravda.ru/1-maya/
(no schema.org properties)

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